

CashSnap Real + Synthetic Data Blend Strategy

Evidence-grounded final version for the next non-overlap model phase

Prepared from the provided CashSnap project context, draft research summaries, and independent source validation.
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Honesty note: This report is intentionally conservative. It does not assume synthetic data helps, does not assume real-only is enough, and does not treat any single metric as proof. The recommendation is to use real data as the anchor and synthetic data as a targeted tool, then prove the recipe against slice-level product scorecards.

Executive Summary

- The right next phase is not more pure synthetic proving. The right next phase is to build the strongest phone-deployable non-overlap CashSnap model using the smartest real + synthetic recipe.
- The current project evidence already says pure synthetic detection found real mechanisms, especially target-vs-unknown pressure, but it is not close enough to product quality. Clean-real controls prove the model/evaluator can reach the target zone, so the remaining problem is data distribution, label semantics, and training objective - not simply model capacity.
- The strongest strategy is real-anchored blending: keep real images as the calibration anchor, use synthetic data only where it covers known missing modes, and require every synthetic addition to beat or protect real-only behavior on slice-level scorecards.
- The first product-level question should be: can synthetic data improve the real model safely? The second should be: if a single 13-class detector cannot handle the tradeoff between recall and unknown/background rejection, does a detector + gate + crop reclassifier pipeline win on count/value behavior within phone/browser constraints?

High-Level Verdicts

Claim	Verdict	Why this is the honest reading
Real + synth is likely better than synth-only.	High confidence.	CashSnap synth-only is currently far below clean-real controls, but synthetic data identified useful mechanisms. The literature generally supports synthetic data as a supplement or pretraining source, not a full replacement.
Real + synth will automatically beat real-only.	Not proven.	Several studies and your own runs show synthetic can hurt if mixed naively or if it teaches artifacts, suppresses target recall, or poisons labels. This must be tested against the strongest real-only control.
Bridged transfer / synthetic pretrain then real calibration is worth testing.	Strong candidate, not a guaranteed answer.	Multiple sources support staged synthetic-to-real transfer, but the exact ratio and schedule are domain-specific.

		Treat as a first experiment, not a conclusion.
Hard negatives / unknown currency are essential.	High confidence, with guardrails.	The CashSnap hardneg6 result and OOD literature both support target-vs-non-target pressure. The risk is recall suppression or mislabeled target negatives.
A two-stage proposal + gate + reclassifier pipeline should be considered.	Strong product candidate.	Project evidence already shows the gate/reclassifier branch can improve count/value behavior. It should compete against the best single-stage real+synth detector, not replace it by assumption.
One headline mAP is enough.	False.	Cash counting needs clean-positive recall, false positives on true empty/unknown, wrong-denomination vs no-box decomposition, count error, value error, source slices, and latency.

CashSnap-Specific Context That Should Drive the Blend

- Final synth-only non-overlap detector is useful but not sufficient: the current verified strict synth-only detector is roughly 0.50 mAP50-95 on the real test set. It is a mechanism result, not a product result. [C1]
- Clean-real controls are much higher, including a near-size real-trained control around 0.82 and a later high clean-real checkpoint around 0.88 on the same evaluator. This argues the gap is mostly data/objective/domain, not raw model capacity. [C1]
- The best product clue is not just single-stage mAP. Strict-best proposals plus a true-empty gate plus a fragment-trained denomination reclassifier produced much stronger real-test product behavior, although the reclassifier uses a small real labeled crop anchor. [C1]
- A recent source-context USD support attempt improved USD source slices but hurt KHR and background behavior. This is exactly why the next phase must use class/source-aware blending and guardrails instead of bulk row dumping. [C1]
- The empty-label bucket is known to be semantically mixed: true empty backgrounds, foreign/currency-review rows, suspect unlabeled targets, and model-review rows must not be treated as the same training role. [C1]

Validation of the Provided Research Drafts

Draft statement / pattern	Validated reading	Action in this final version
"Synthetic is supplementary; real data should anchor the model."	Supported by project evidence and multiple synthetic-data studies.	Kept as a core principle.
"Pretrain on synthetic then fine-tune on real."	Supported as a strong candidate, especially when real data is limited; not guaranteed to beat real-only in this domain.	Kept as a priority experiment, not a claim of optimality.
Hard-coded ratios like 25-50% synth, 70% real, 100K synth / 5K real.	Potentially useful starting points, but too specific without CashSnap evidence.	Replaced with ratio/schedule sweeps and kill conditions.
"Unknown class" as a direct detector class.	The general idea is valid, but CashSnap direct 14-class UNKNOWN experiments failed. Unknown handling is better tested via a gate/crop reject path first.	Demoted from default recipe to conditional architecture branch.
"Two-stage hybrid pipeline is optimal."	Promising because CashSnap has wrong-	Kept as a product candidate, not as a

	denomination and unknown-rejection issues, but edge latency and real product metrics must prove it.	final assumption.
Specific latency/FLOP claims for YOLO variants.	Not reliable enough without local phone/browser benchmarks.	Removed as factual claims; added a requirement to benchmark actual exported models.
Blogs/forums/travel pages as sources.	Useful for intuition only, weak as evidence.	Excluded from core recommendations unless backed by stronger papers or project results.

Principles for the Best Real + Synthetic Blend

- 1. Real images are the calibration anchor.** The final model will be deployed on real phone images. Synthetic rows are allowed only if they improve or protect real behavior. Hold out real-only validation/test slices and do not evaluate on synthetic.
- 2. Synthetic rows should have a job.** Do not add synthetic data because it is cheap. Each synthetic source should map to a known failure: weak class, weak source, missing extent, camera/lighting gap, unknown rejection, background overfire, or crop denomination confusion.
- 3. Stage or sample; do not blindly mix.** Staged training and carefully balanced sampling are safer than unweighted pooling. A model can learn synthetic artifacts if synthetic dominates too late in training.
- 4. Separate target recall from rejection pressure.** Hard negatives are powerful, but they can suppress true banknotes. Use separate guardrails for target recall, true-empty false positives, foreign/unknown false positives, and value/count accuracy.
- 5. Label semantics are part of the model.** A zero-label true-empty image, a foreign banknote, an unlabeled USD/KHR bill, and an ambiguous sliver are not the same object. Treating them the same is label poison.
- 6. Use product metrics to arbitrate.** For a cash counter, wrong value can matter more than a box IoU. A model that slightly improves mAP but worsens value error or background hallucinations is not better.
- 7. Architecture is a data strategy.** If a single detector cannot balance localization, denomination, and OOD rejection, the correct data blend may include a proposal detector, reject gate, and crop classifier.

Required Data Taxonomy

Bucket	Training role	Risk if mishandled
Target USD/KHR notes	Positive labels for current core classes. Include front/back, scale, lighting, condition, source diversity.	Unlabeled target notes in negatives teach the model not to detect real money.
Clean/true empty backgrounds	Zero-label negatives for background suppression.	If overused or too easy, may not fix banknote-like FPs; if mixed with targets, poisons recall.
Foreign/unknown currency	Hard rejection pressure. Prefer curated zero-label negatives for single-stage detector, or reject/unknown labels for gate/classifier.	Can suppress target recall if too close to target classes or treated too bluntly.
Receipts, paper props, cards, packaging	Non-banknote hard negatives, especially	Can improve background precision but

	for retail clutter.	may not solve foreign-banknote confusion.
Suspect unlabeled USD/KHR	Review/manual label/ignore. Never use as zero-label background.	Direct label poison.
Ambiguous slivers / human-unidentifiable fragments	Ignore/review bucket; for crop classifier, may be reject examples if clearly non-actionable.	Forced denomination labels create noisy class boundaries.
Synthetic positives	Targeted support for known missing modes: weak denominations, extent, lighting, source, camera variation.	Pretty but wrong-context rows can worsen real transfer.
Synthetic unknowns / distractors	Controlled rejection pressure, ideally with paired target examples and source balancing.	Direct unknown classes can become a new false-positive output if not evaluated carefully.

Recommended Data Strategy for the Next Phase

1. Freeze a strong real-only baseline and a strong real-only validation protocol. This is the anchor. Any real+synth recipe must compare against it, not against weak synthetic-only baselines.
2. Clean the evaluation semantics before declaring wins. Maintain separate slices for clean positives, true empty, foreign/unknown currency, suspect unlabeled targets, source-specific subsets, and clutter/lighting stress.
3. Build a controlled real+synth blend harness. It must support staged training, per-source sampling weights, per-class quotas, hard-negative caps, train-phase parity checks, and product scorecards.
4. Test synthetic as targeted interventions: weak USD/source repair, KHR protection, foreign/unknown rejection, background overfire, visible-extent/localization repair, and crop denomination support.
5. Prefer small, interpretable blends first. If a small targeted synth set cannot improve the real baseline on its intended slice, a much larger version is unlikely to be the right path without a new mechanism.
6. Promote only recipes that improve product behavior. A recipe can be kept as a clue if it improves one slice but hurts another; it should not become the new baseline unless the tradeoff is acceptable.

Training Recipes to Compare

Recipe	What it tests	Promotion logic
A. Real-only control	The actual bar to beat.	Must be maintained and re-run when harness changes.
B. Synthetic pretrain -> real fine-tune	Whether synthetic creates useful reusable features before real calibration.	Promote if it beats real-only on full scorecard without class/background regressions.
C. Real pretrain -> targeted synth repair -> real calibration	Whether synthetic can repair specific weak modes after real features are learned.	Promote only if final real calibration preserves gains and prevents synth artifacts.
D. Balanced mixed batches	Whether simultaneous real+synth sampling helps regularization or source balance.	Kill if it underperforms staged training or causes synthetic artifact leakage.
E. Class/source-aware sampling	Whether rare/weak denominations and source modes need controlled gradients rather than more rows.	Promote if it repairs weak slices without hurting protected KHR/high-value classes.
F. Detector + gate + crop reclassifier	Whether factorizing localization, rejection, and denomination solves conflicts single-stage training cannot.	Promote if count/value error and unknown/background behavior beat single-stage within latency budget.

Evaluation Scorecard: Do Not Use One Metric Alone

Metric / slice	Question answered	Why it matters
Full real mAP50-95	Does the detector improve overall?	Useful but can hide class and background failures.
Per-class AP / recall	Which denominations are harmed?	A cash app cannot trade away high-value or rare notes silently.
Clean-positive slice	Does it detect clear non-overlap real notes?	This is the foundation before overlap/fan/hand data.
True-empty FP images / detections	Does it hallucinate money on no-note scenes?	False money detections break user trust.
Foreign/unknown currency FP rate	Does it reject non-target notes?	Critical in mixed retail/travel environments.
Wrong-denomination vs no-box FNs	Is the issue localization or classification?	Determines whether detector data or crop classifier data is the next bottleneck.
Source slices	Which source/camera/context modes fail?	Prevents a blend from improving one source while destroying another.
Count MAE / exact count	Does the app count notes correctly?	Direct product metric.
Value MAE / exact value	Does it count the money correctly?	Wrong \$50 vs \$20 can matter more than a minor IoU issue.
Latency/model size	Can it run on phone/browser?	A slightly better but too-slow model is not the end goal.

Top Experiments to Run Next

Rank	Experiment	What would make it a real win	Kill condition
1	Real-only baseline + slice harness refresh	Confirms the bar, exposes real weak slices, and prevents fake wins.	If labels/slices are too messy to trust, stop training experiments and fix the eval bridge first.
2	Staged blend: synthetic pretrain -> real fine-tune vs real-only vs naive mix	A real+synth recipe beats real-only on mAP and product metrics without class/background regressions.	If gains disappear after real calibration or only beat weak baselines, keep lesson but do not promote.
3	Hard-negative/unknown strategy under real anchor	Foreign/unknown and true-empty FPs fall while real target recall stays flat or improves.	Kill ratios/objectives that suppress protected classes or target recall.
4	Class/source-aware synthetic repair	Weak USD/source modes improve without KHR/source regressions.	Kill if it repeats the known USD-up/KHR-down/background-worse tradeoff.
5	Product stack bakeoff: single-stage detector vs detector + gate + crop reclassifier	Count/value behavior improves within mobile/browser latency and maintainability limits.	Kill if latency or complexity rises without meaningful value/count improvement.
6	Active learning / manual label bridge	Small new real labels fix high-uncertainty, high-value failure modes faster than synthetic tweaks.	Kill if selected samples are redundant or do not move the scorecard.

Suggested First Two Weeks

- Day 1-2: Freeze real-only control, synth-only reference, and current product stack. Re-run only if cache/train-phase issues are suspected. Build one dashboard that compares all scorecard slices.

- Day 3-4: Audit real eval/train semantics. Mark true-empty, foreign/unknown, suspect target, ambiguous/ignore, and clean-positive slices. Do not train on unverified empty-label rows as negatives.
- Day 5-7: Run the three core blend baselines: real-only, naive real+synth mix, and synthetic pretrain -> real fine-tune. Keep budgets and train phases comparable.
- Week 2: Run targeted interventions only where the dashboard shows a real bottleneck: hard negatives for FPs, source-aware synthetic for USD/source misses, KHR-protection sampling, or crop/gate product stack improvements.
- End of week 2: Choose the best non-overlap recipe by product scorecard, not by aggregate mAP alone. Archive failures with the mechanism learned.

Rabbit Holes to Avoid

- Bulk synthetic row count without a failure target. More rows will not fix label semantics or a wrong objective.
- Pretty contact sheets as proof. Visual plausibility is a gate, not an outcome metric.
- Hard-negative dose sweeps without separating true empty, foreign/unknown, and suspect target rows.
- Direct UNKNOWN class scaling unless the architecture and scorecard explain why earlier direct unknown attempts failed.
- Ultra-realistic renderers or diffusion refiners before a small controlled transfer test proves they help real scorecards.
- Optimizing only mAP while ignoring count/value error, false positives, and weak class regressions.
- Treating source-context replacement as solved just because teacher agreement is high. It must still beat real transfer and source slices.
- Overbuilding architecture before benchmarking export latency and testing whether the product metrics actually improve.

Research Evidence: What It Really Supports

Evidence area	Supported conclusion	How CashSnap should use it
Synthetic + real object detection studies	Synthetic can help, especially as supplement/pretraining, but naive mixing can underperform due to domain gaps.	Test staged and balanced recipes against real-only; do not assume the ratio.
Bridged transfer / style alignment	Synthetic pre-adaptation followed by real adaptation is plausible and often stronger than naive blending.	Make synthetic pretrain -> real fine-tune a first-class baseline.
Domain randomization	Broad variation can help transfer when the randomized factors cover real variation.	Randomize camera/lighting/blur/context only if it maps to real failure slices.
Cut-paste synthesis	Patch-level realism and artifact control can be enough for detectors; random/context-wrong paste can hurt.	Use source/context accounting and artifact audits for banknote composites.
Hard negatives / OOD	Negative information near the decision boundary is valuable, but unverified negatives can poison recall.	Curate true negatives and evaluate recall/FP tradeoffs separately.
Active learning	Uncertainty + diversity selection is a good way to spend real-label budget.	Label the real examples that most reduce uncertainty in weak classes/sources.
Currency detection papers	Mobile currency recognition can work, including mixed and partially covered	Use as deployment inspiration, not as proof CashSnap is solved.

	currencies, but most results are dataset-specific.	
Generative metrics vs detection	Global image metrics may not predict YOLO performance.	Judge synthetic data by downstream CashSnap scorecards, not FID/contact-sheet beauty.

Recommended Target End State for Non-Overlap

- A compact single-stage detector remains the simplest baseline, but the best product candidate should be selected by bakeoff, not preference.
- If the two-stage pipeline keeps winning count/value and rejection metrics, the final non-overlap foundation should be a small detector plus a lightweight reject/gate/classifier path.
- The winning recipe should include a documented data blend: real anchor set, synthetic sources used, negative/unknown policy, sampling schedule, and what each data bucket was intended to fix.
- The final non-overlap model should include a short model card: intended classes, non-target behavior, weak slices, latency, count/value metrics, and what overlap/fan/hand cases are not yet covered.

Source Validation Notes

Source type	Use level	Notes
Peer-reviewed / conference / arXiv papers	Primary evidence	Used for broad principles: synthetic transfer, domain randomization, cut-paste synthesis, active learning, OOD/negative prompts.
Official project model.md	Primary project evidence	Used for CashSnap current state, tested/rejected paths, and known failure modes.
Official National Bank of Cambodia banknote page	Authoritative currency reference	Use for KHR denominations/design scope checks, but CashSnap class scope remains the project-defined core set.
Ultralytics docs/issues/forums	Practical implementation evidence	Useful for YOLO behavior around empty-label images and deployment/export, but not a scientific proof of optimal ratios.
Blogs, Reddit, travel sites, ResearchGate mirrors, Moonlight/EmergentMind summaries	Low-confidence / supplemental only	Do not base final technical claims on these unless backed by primary sources.
Provided draft PDF and pasted blueprint	Input drafts, not evidence by themselves	Used to identify candidate claims; over-specific numbers and unsupported latency claims were corrected or removed.

References

[C1] CashSnap model.md, uploaded project brain, June 2026.

[D1] Executive Summary.pdf, uploaded draft research summary.

[D2] Pasted text(13).txt, uploaded technical blueprint draft.

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